

CFD Assignment

Assignment for AE4202 – CFD for Aerospace Engineers

Group Assignment – November 2025

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by

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Introduction

This report presents a computational fluid dynamics analysis of the NACA0012 airfoil using 2D steady-state Reynolds-Averaged Navier-Stokes simulations. The aerodynamic performance of this symmetric airfoil will be investigated at three angles of attack: 0° , 10° and 15° , at a Reynolds number of chord-length of 6 million and a free-stream Mach number of 0.30.

The simulations will be conducted under ambient conditions of 250 K total temperature and 50 kPa total pressure, with air modeled as a compressible ideal gas following Sutherland's law for dynamic viscosity. The airfoil geometry will be scaled accordingly to meet the Mach value of 0.3 at the inlet.

The workflow begins with the generation and analysis of two structured multi-block meshes: an initial coarse mesh (Mesh 1) with fewer than 20,000 cells, and a refined mesh (Mesh 2) designed to achieve y^+ values below one for wall-resolved turbulence modeling. The $k-\omega$ turbulence model will be used as the baseline model due to its performance in resolving near-wall flows and adverse pressure gradients.

Finally, verification of the numerical solution will be performed by evaluating the sensitivity to spatial discretizations schemes and turbulence modeling approaches. All the results will be validated against the experimental reference data provided for the pressure distributions along the airfoil surface.

1 Scaling Airfoil Parameters

Given a chord-length Reynolds number 6 million, free-stream Mach number 0.30, ambient conditions of air with total temperature of $250K$ and total pressure $50kPa$, the air is modeled as a compressible ideal gas with a dynamic viscosity given by Sutherland's law.

In order to find the airfoil's chord length, first the isentropic temperature and pressure at the given Mach number is calculated using Equation 1.1 and Equation 1.2.

$$T = \frac{T_0}{1 + \frac{\gamma-1}{2} M^2} = 245.58 Pa \quad (1.1)$$

$$P = \frac{P_0}{\left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma}{\gamma-1}}} = 46.97 kPa \quad (1.2)$$

The free stream density can therefore be calculated using Equation 1.3.

$$\rho = \frac{P}{RT} = 0.666 kg m^{-3} \quad (1.3)$$

The free stream velocity can also be found by first calculating the speed of sound with Equation 1.4 and then with Equation 1.5.

$$a = \sqrt{\gamma RT} = 314.12 \quad (1.4)$$

$$V = M a = 94.24 m s^{-1} \quad (1.5)$$

Furthermore, the dynamic viscosity is calculated using Sutherland's law as shown in Equation 1.6, with the reference viscosity of $1.716 \cdot 10^{-5}$ and S being equal to $110.4 K$.

$$\mu(T) = \mu_0 \left(\frac{T}{T_0} \right)^{\frac{3}{2}} \frac{T_0 + S}{T + S} = 1.576 \cdot 10^{-5} \quad (1.6)$$

Finally, the chord length is found by rearranging the equation that defined the Reynolds number as given by Equation 1.7.

$$Re_c = \frac{\rho U c}{\mu} \quad (1.7)$$

$$c = \frac{Re_c \mu}{\rho U} = 1.5057 m \quad (1.8)$$

2 Mesh

First the airfoil is opened in ICEM CFD, and then it is scaled up to the previously calculated chord length of $1.5057m$, as seen in Figure 2.1. A computational domain is created consisting of a semi-circular front section and two tangent lines connecting to an aft vertical boundary enclosing the airfoil. The domain extends approximately ten airfoil lengths upstream from the leading edge, twenty airfoil lengths downstream, and fifteen airfoil lengths above and below the symmetry line. The geometry shape is then transformed into parabolic inlet shape. The final computational domain can be seen in Figure 2.2.

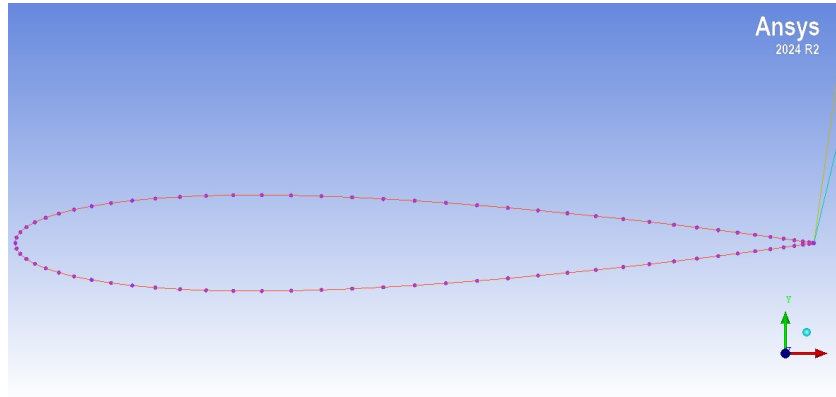


Figure 2.1: Scaled Airfoil Geometry

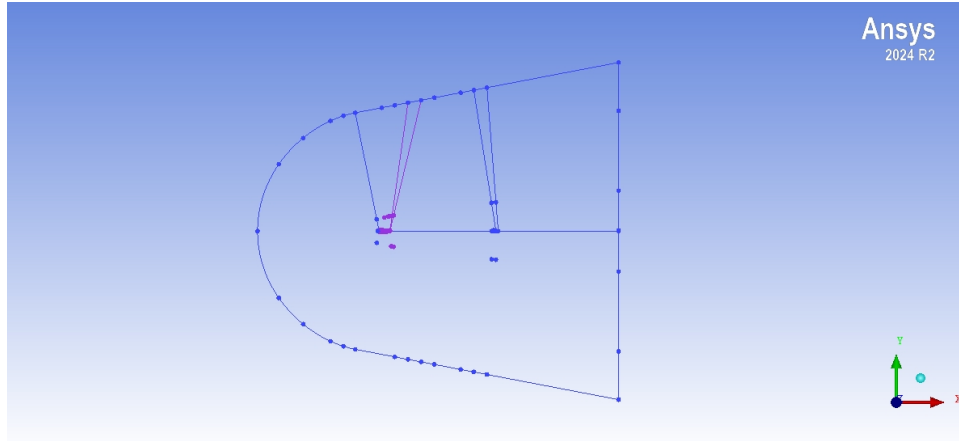


Figure 2.2: Grid 1: Geometry Domain

Then, in order to create a structured mesh, the geometry has to be divided into blocks. The geometry is first extruded into the Z-axis by $1m$ to transform the geometry from a 2D into a 3D shape, and then the airfoil surface is cut out from the domain. Then 5 parts are created and matched to the surfaces: the inlet, outlet, airfoil, and both symmetry faces named positive and negative symmetries.

Then one big block is generated and split to get a C-grid with the o-grid option and selecting both symmetry and outlet faces. Some additional points are added to the domain, also extruded in the Z-axis, which are then used to split the block into pieces by associating vertices with these additional points. The final blocking can be seen in Figure 2.3.

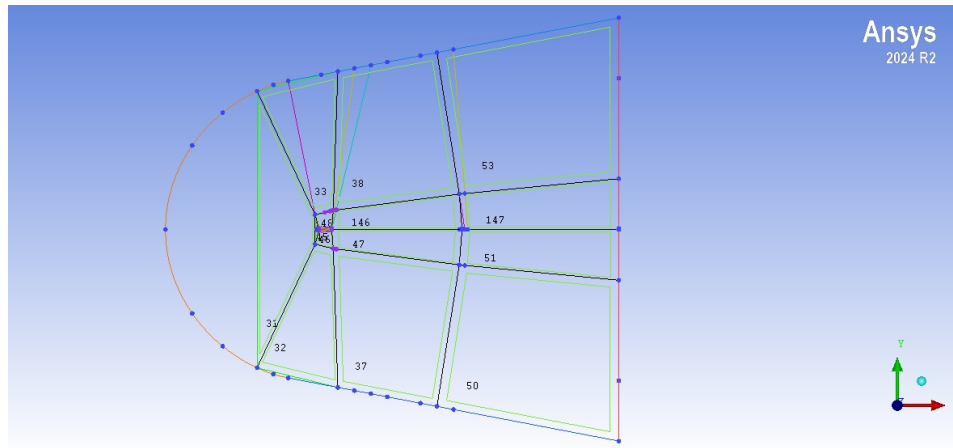


Figure 2.3: Final Blocking Grid 1

Next, the mesh characteristics are selected. A radial discretizations with an exponential spacing is applied, making the spaces wider as they approach the geometry boundaries. For the spacing along the airfoil geometry, several spacing distributions were tried out until a good enough quality was achieved. This meant that the volume and determinant was bigger than zero, and the minimum angle was greater than thirty degrees. The final mesh and its quality is displayed below, with 19845 cells.

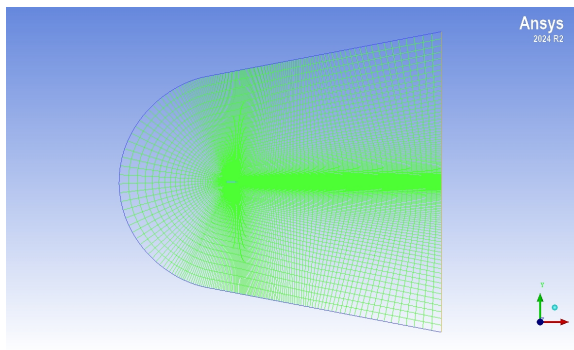


Figure 2.4: Mesh 1

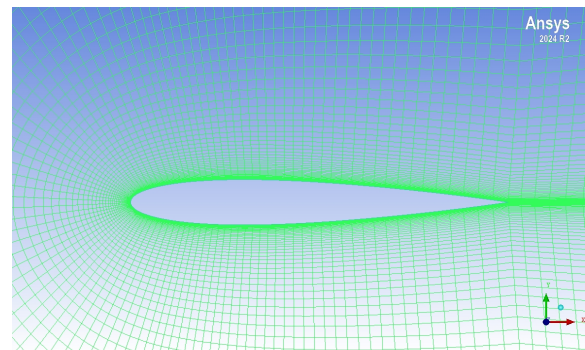


Figure 2.5: Airfoil Mesh 1

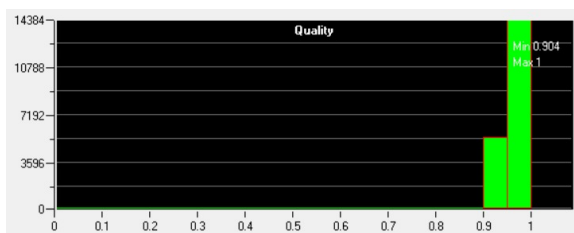


Figure 2.6: Quality of Mesh 1

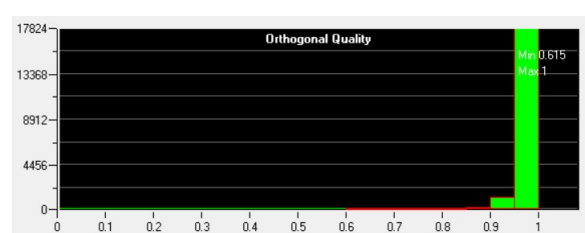


Figure 2.7: Orthogonality of Mesh 1

3 Steady State Simulation for Grid 1

3.1. Boundary Conditions

Once the computational domain is meshed, it can be exported and the simulation can be set-up using CFX-Pre. In the Pre-Solver the boundary conditions at each part is defined.

For the inlet, the following parameters are chosen: a subsonic flow regime, with a total relative pressure of $50000Pa$, total temperature of $250K$, and an X component flow direction of $94.24ms^{-1}$, as calculated in Equation 1.5. The turbulence at the inlet is set to have a fractional intensity of 0.03, as given in the assignment, and an eddy length scale of $0.15m$. This value was calculating my multiplying the airfoil's chord length by an approximated turbulence length scale of 0.1.

For the outlet, only the flow regime is specified to subsonic and the static pressure equal to $46.97kPa$ from Equation 1.2.

Boundary conditions are also imposed at the airfoil. These include a no slip smooth wall condition at the airfoil's surface and an adiabatic heat transfer.

Finally, a symmetry boundary condition is imposed on both symmetry faces.

Then some simulation setup is specified. The flow material is specified to be ideal gas air and enabling compressible ideal gas equation of state. Sutherland's law is chosen as the viscosity model with a reference viscosity of $1.716e^{-5}Pa \cdot s$, reference temperature of $273.15K$, temperature exponent of 1.5 and a Sutherland constant of $110.4K$ so that viscosity varies with temperature in the solution.

At the solver control, the advection and numerical scheme is set to high resolution. A maximum of 3000 iterations is specified for most of the cases, just to ensure that residuals stay stable but in most of them there is no need for that many iterations, with an automatic timescale and conservative length scale.

3.2. Results

Once the boundary conditions are imposed, the mesh is run in the solver.

3.2.1. Solver Residuals

The solver residuals are plotted in Figure 3.1. The plot verifies the convergence of the simulation as the convergence value is below $1 \cdot 10^{-6}$. The force convergence is shown in Figure 3.2.

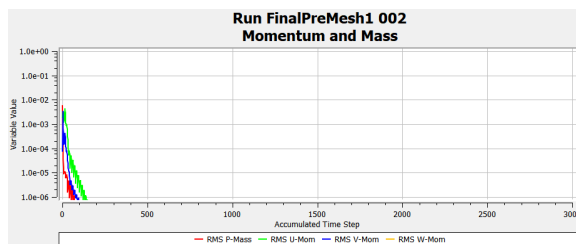


Figure 3.1: Solver Residuals

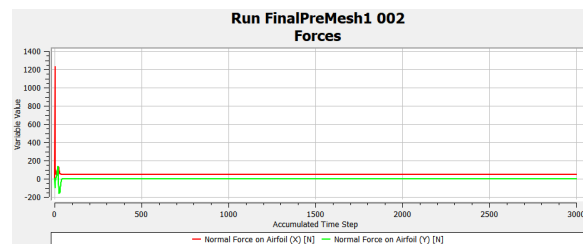


Figure 3.2: Force Convergence Grid 1

3.2.2. Flow Field

Finally, the simulation results can be exported into the Post-Solver. The contour plots for static pressure and velocity are shown in Figure 3.3 and Figure 3.4 respectively.

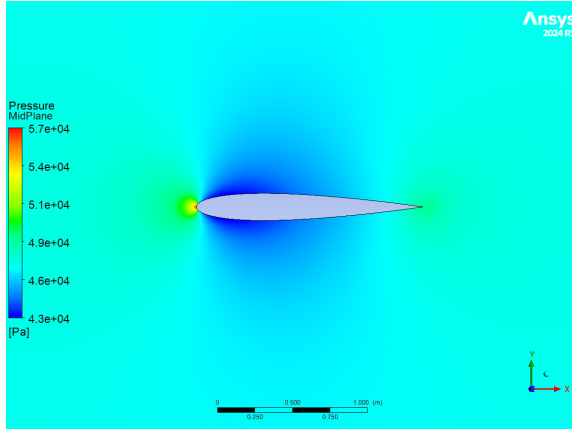


Figure 3.3: Static Pressure Contour Plot for Grid 1

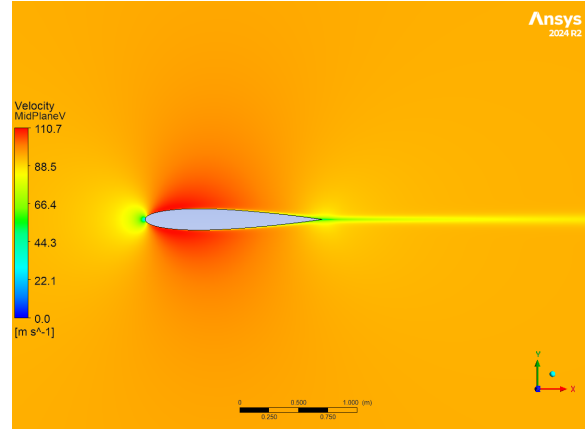


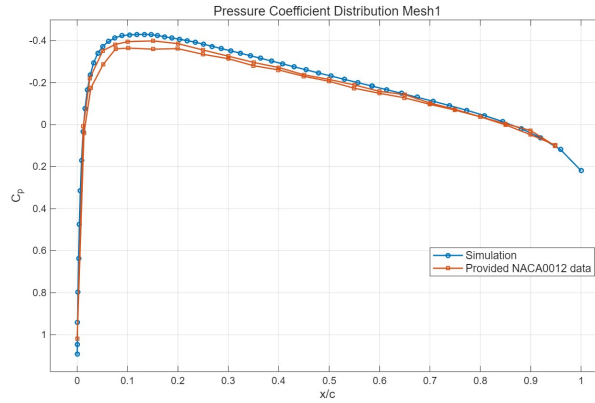
Figure 3.4: Velocity Contour Plot for Grid 1

3.2.3. Non-Dimensional Numbers

The non-dimensional pressure coefficient C_p , using

$$C_p = \frac{p - p_\infty}{\frac{1}{2} \rho_\infty V_\infty^2} \quad (3.1)$$

along the airfoil surface, together with the provided reference data is plotted in Figure 3.5.

Figure 3.5: C_p Distribution of Airfoil and Reference Data

From the graph it can be seen that the simulation provides a smoother distribution and the C_p for both the upper and lower surfaces coincides, which is the expected outcome of a symmetrical airfoil at zero angle of attack.

The non-dimensional lift and drag coefficient and the maximum Mach number for Grid 1 is summarized in Table 3.1.

Table 3.1: Non-Dimensional Coefficients of Grid 1 and Reference Data

Geometry	M_{max}	C_D	C_L
Airfoil Grid 1	0.3534	9.5162e-03	-9.7962e-06

The lift coefficient should be zero for a symmetrical airfoil at zero angle of attack, as the airflow distribution over the upper and lower surfaces are identical. The very small non-zero value of C_L given on the table is due to numerical round-off and minor mesh asymmetries. The small but finite drag coefficient corresponds to viscous drag effects. Thus, the results matches the expected physical behavior, although the mesh could be improved to reduce mesh asymmetries.

Finally, the y^+ value of Grid 1 airfoil is plotted in Figure 3.6. The y^+ value is defined as

$$y^+ = \frac{u_\tau y}{\nu} \quad (3.2)$$

and, as seen from the plot, the y^+ value varies from 30 up to 280. This means that grid has to be refined and improved to achieve a y^+ lower than 1 because, as previously stated, a $k-\omega$ model is used which is designed to resolve the boundary layer all the way down to the wall. This is to ensure that the velocity gradient and shear stress near the wall are computed accurately from the physics, not approximated. Therefore a finer grid is generated called Grid 2.

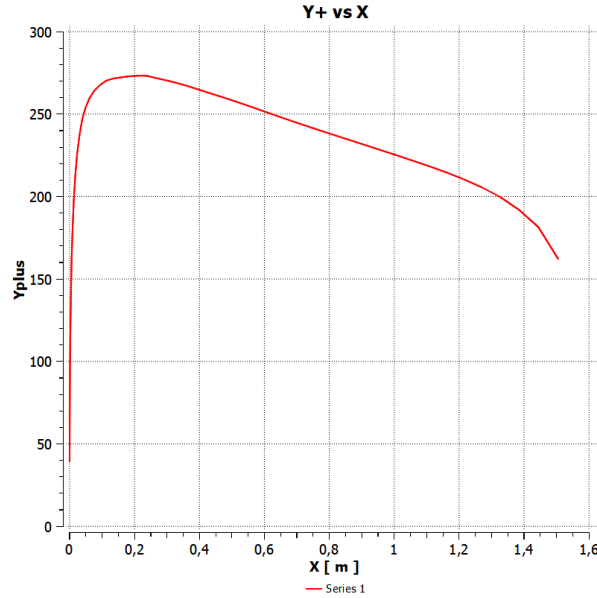


Figure 3.6: y^+ Value of Grid 1

4 Refinement of Grid - Grid 2

Grid 2 was refined by first increasing the number of nodes along the airfoil surface. The number of cells is increased to 57680. Then, we adjusted exponential spacing ratios at the boundaries to ensure smooth cell transitions while also maintaining good orthogonality throughout the whole domain. Also, the first layer along the airfoil was reduced to around $2 \cdot 10^{-6}$ to achieve the target $y^+ < 1$. The finer mesh can be seen in Figure 4.1 and Figure 4.2. Furthermore, Figure 4.4 shows the new mesh quality. This mesh could be further improved by refining more the mesh and increasing the quality and orthogonality to get more precise results and faster convergence.

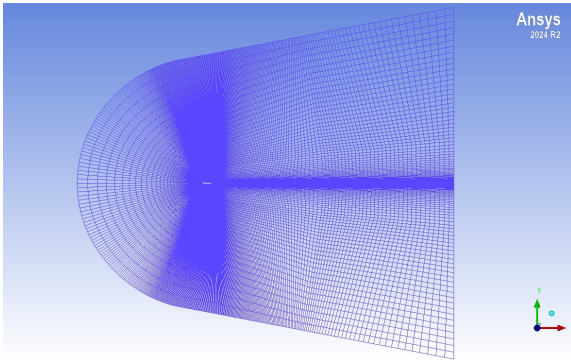


Figure 4.1: Mesh 2

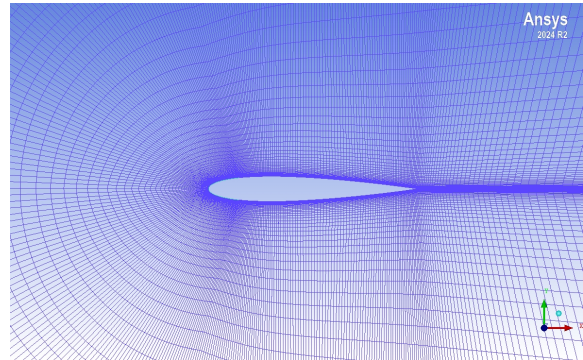


Figure 4.2: Airfoil Mesh 2

From Figure 4.3 it can be clearly verified that the target solution has been achieved at the y^+ value is below 1 along the entire chord.

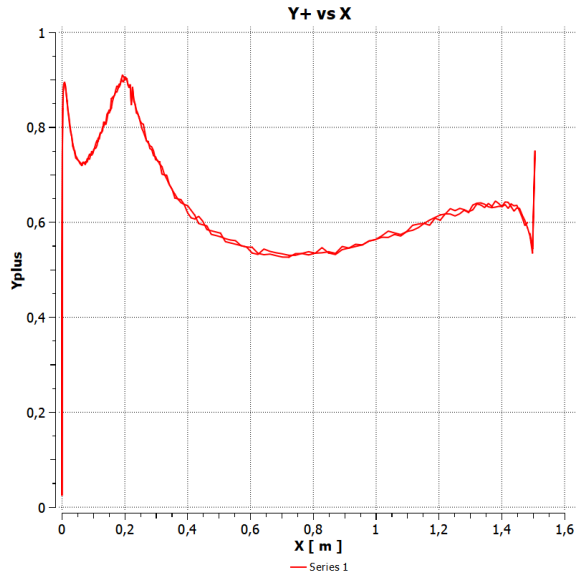


Figure 4.3: y^+ Value of Grid 2 at 0°

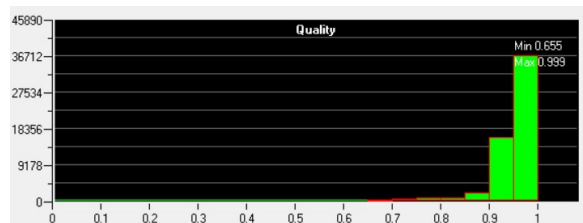


Figure 4.4: Quality Mesh 2

4.1. Results Grid 2

4.1.1. Solver Residual

The residuals for this finer mesh are shown in Figure 4.5. They decrease rapidly during the first iterations and stabilize at very low values, below $1 \cdot 10^{-4}$, indicating that the solution has converged to a steady state. The final values only fluctuate slightly around a stable baseline suggesting that no further changes occur in the flow field. The following figure, Figure 4.6, demonstrates the convergence of the forces on both the X-axis (Lift) and Y-axis (Drag).

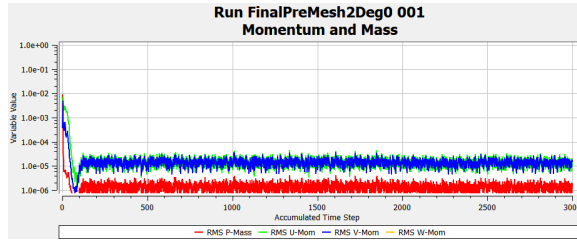


Figure 4.5: Solver Residuals for Grid 2

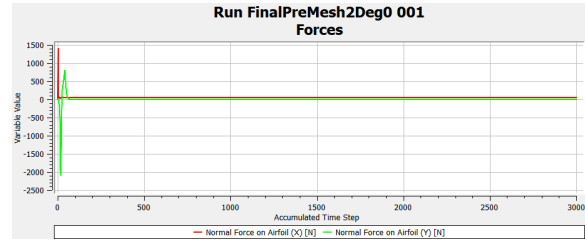


Figure 4.6: Force Convergence for Grid 2

4.1.2. Flow Field

Both the static pressure and velocity contour plots for the refined mesh are shown in Figure 4.7 and Figure 4.8 respectively. Both contour plots are almost identical to the ones from the coarser mesh. For both cases the symmetrical airfoil at zero angle of attack has low suction peaks so the finer mesh cannot really display significant differences.

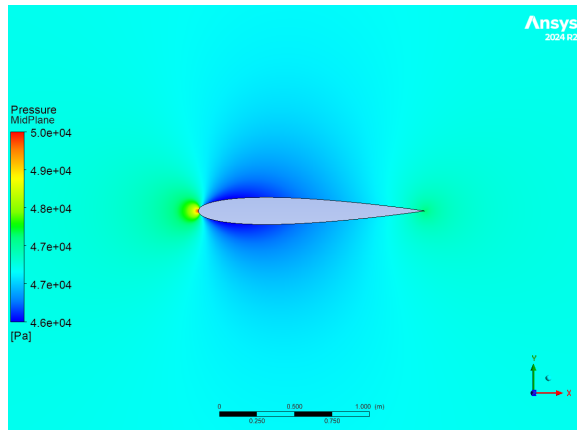


Figure 4.7: Static Pressure Contour Plot for Grid 2

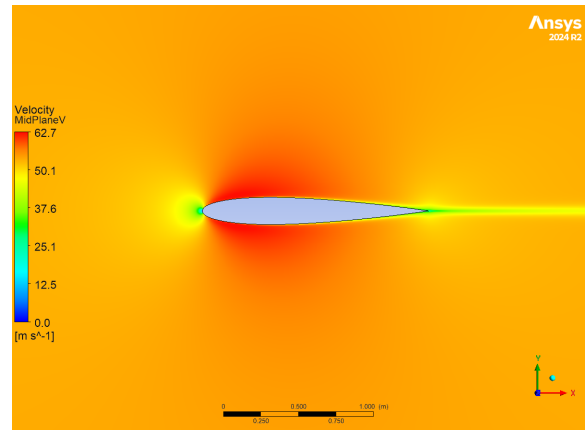


Figure 4.8: Velocity Contour Plot for Grid 2

4.1.3. Non-Dimensional Numbers

The C_p distribution obtained from the finer grid is displayed in Figure 4.9. It contains a higher number of data points because the refined mesh includes more surface nodes along the airfoil. This increases spatial resolution which results in a smoother and more detailed plot, specially near the leading edge where the pressure gradient is the strongest. The overall shape and magnitude of the pressure coefficient remains nearly identical between both meshes, indicating that the solution has reached grid independence.

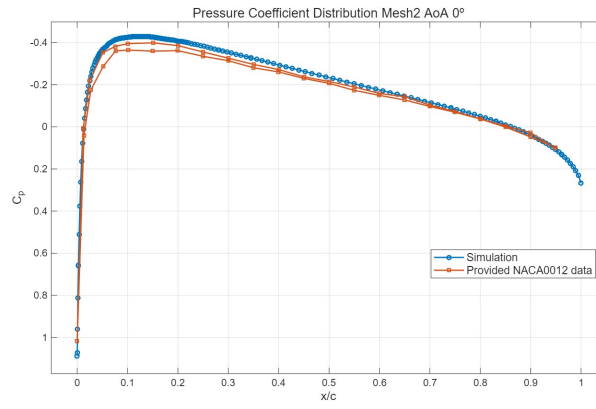


Figure 4.9: *C_p distribution of Airfoil Grid 2 and Reference Data*

The new non-dimensional numbers are summarized in Table 4.1. The values are very close to the ones from the coarser mesh, however, the magnitude of C_L is closer to zero and the C_D value is slightly increased due the mesh refinement.

Table 4.1: *Non-Dimensional Coefficients of Grid 2 and Reference Data*

Geometry	M_{max}	C_D	C_L
Airfoil Grid 2	0.3530	9.5792e-03	5.2556e-06

5 Grid Rotation

The fine mesh Grid 2 is now simulated for two other angles of attack, 10° and 15°. In order to achieve this, the flow velocity at the inlet has an X and Y component. Therefore, the lift and drag forces were expressed by projecting the total aerodynamic forces perpendicular and parallel to the freestream velocity vector. The equations for lift and drag are given by Equation 5.1 and Equation 5.2 respectively.

$$L = -F_x \frac{V_y}{\sqrt{V_x^2 + V_y^2}} + F_y \frac{V_x}{\sqrt{V_x^2 + V_y^2}} \quad (5.1)$$

$$D = F_x \frac{V_x}{\sqrt{V_x^2 + V_y^2}} + F_y \frac{V_y}{\sqrt{V_x^2 + V_y^2}} \quad (5.2)$$

5.1. Results Grid 2 10° Angle of Attack

For the 10° angle of attack simulation, $V_x = V \cos 10$ and $V_y = V \sin 10$.

5.1.1. Solver Residual

Figure 5.1 shows the solver residuals. As before, all residuals decrease sharply during the initial iterations before stabilizing around constant values indicating a steady state solution, confirming a numerical stable and converged solution. The residuals for all variables remain below 10^{-3} and most of them below 10^{-4} , which is acceptable for this angle of attack where some flow separation can make the convergence slightly slower. Figure 5.2 presents the convergence of the forces. Both of them also converge.

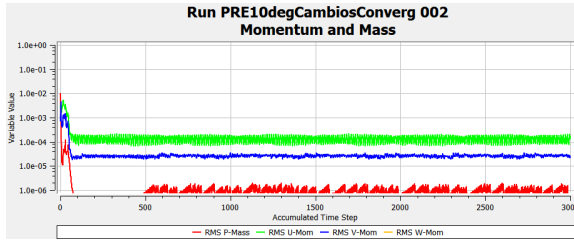


Figure 5.1: Solver Residuals for Grid 2 at 10°

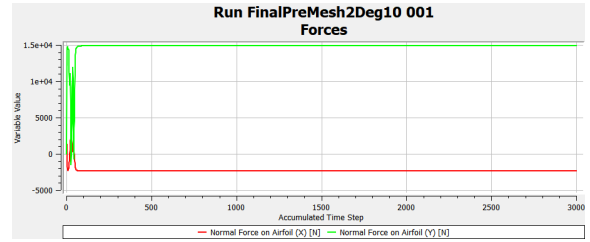


Figure 5.2: Force Convergence for Grid 2 at 10°

5.1.2. Flow Field

From Figure 5.3, it can be seen that a low-pressure region has been formed over the upper surface due to flow acceleration, while a high-pressure region has been formed beneath the airfoil. This pressure difference now generates a lift force acting on the airfoil. The leading edge suction peak is now visible (the greener area). The velocity contour plot, displayed in Figure 5.4, mirrors this behavior. It can be seen a higher velocity over the airfoil. Also, the drag generation can be spotted from the wake region downstream of the trailing edge.

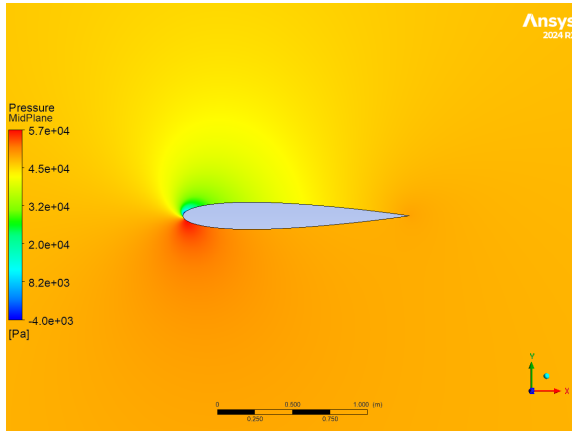


Figure 5.3: Static Pressure Contour Plot for Grid 2 at 10°

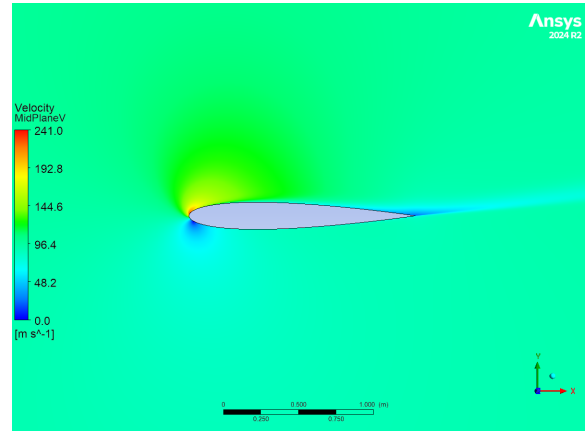


Figure 5.4: Velocity Contour Plot for Grid 2 at 10°

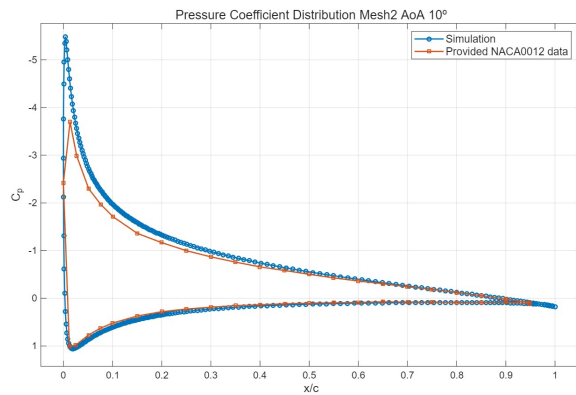
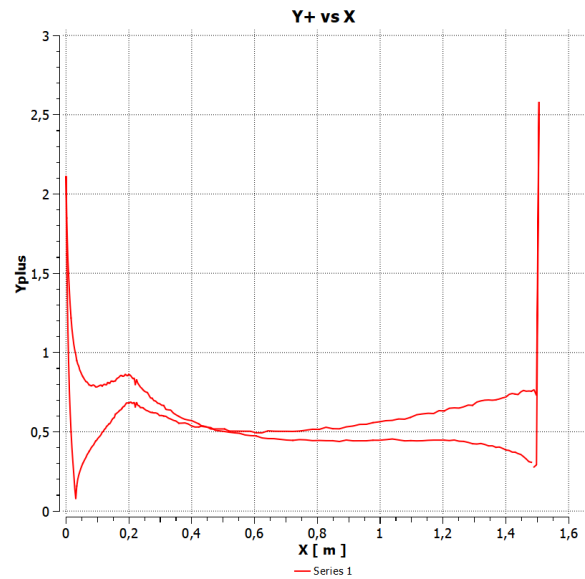
5.1.3. Non-Dimensional Numbers

The new non-dimensional numbers, summarized in Table 5.1, reflect the changes in the flow fields. The lift coefficient has increased from almost zero to a positive value, matching the expectations for an airfoil at positive incidence angle. Consequently, drag coefficient also increases due to increased flow curvature and viscous losses. Additionally, the Mach number grows from 0.35 to 0.81, indicating stronger flow acceleration over the upper surface due to the suction peak.

Table 5.1: Non-Dimensional Coefficients of Grid 2 at 10°

Geometry	M_{max}	C_D	C_L
Airfoil Grid 2 at 10°	0.8091	2.5754e-02	1.0795

This increase in suction peak clearly seen in Figure 5.5. The C_p distribution is now asymmetric, proving the generation of lift discussed above. The suction peak is lower for the reference data, indicating that the model over-predicts the higher flow acceleration near the leading edge. However, this can result from the reference data points being discrete and sparse around the leading edge, meaning that the interpolation line does not completely restore the true suction peak. Nevertheless, there is correlation with the reference data throughout the rest of the chord-wise pressure distribution that validates the numerical model.

Figure 5.5: C_p Distribution of Airfoil Grid 2 and Reference Data at 10°Figure 5.6: y^+ Distribution of Grid 2 at 10°

Finally, Figure 5.6 shows the y^+ plot for 10°. The plot shows peaks at the leading and trailing edge suggesting higher wall shear stress and velocity gradients in those regions, as the cell heights are unchanged. The y^+ value for the rest of the chord length is below 1 which is valid for the wall-resolved simulation.

5.2. Results Grid 2 15° Angle of Attack

For the 10° angle of attack simulation, $V_x = V \cos 15$ and $V_y = V \sin 15$.

5.2.1. Solver Residual

Figure 5.9 shows the RMS residuals throughout the simulation. Although the residual levels are slightly higher than those obtained at the other two lower angles of attack, this is expected because the flow at 15° involves stronger flow separation. Despite this, the residuals remain below 10^{-3} – 10^{-4} indicating a numerically stable steady state solution. Figure 5.8 also shows the converge of both forces acting on the airfoil.

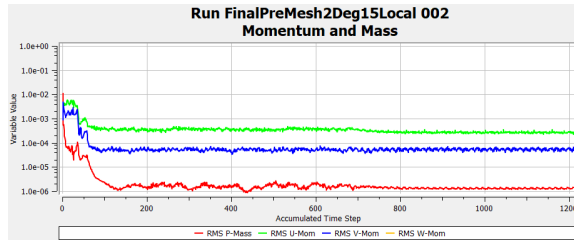


Figure 5.7: Solver Residuals for Grid 2 at 15°

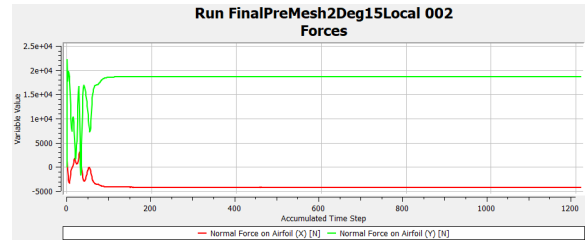


Figure 5.8: Force Convergence for Grid 2 at 15°

5.2.2. Flow Field

Figure 5.9 and Figure 5.10 show the static pressure and the velocity contour plots. The plots clearly show stronger pressure gradients and larger flow separation compared to the case of 10°. The pressure on both the upper and lower side has increased, showing a more pronounced pressure difference and also flow separation over the upper surface. This is confirmed by the velocity contour plot. Flow separation occurs earlier and the wake behind the trailing edge is larger than before.

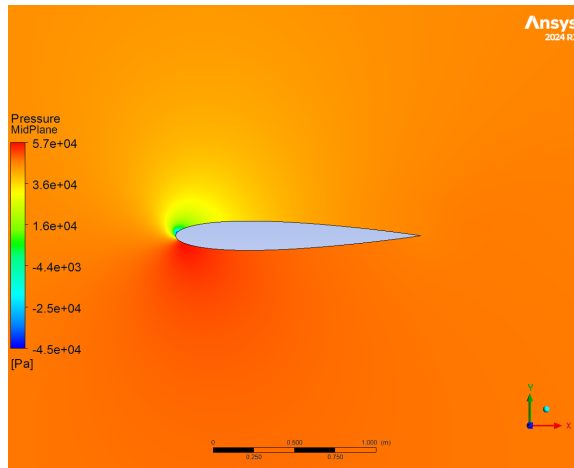


Figure 5.9: Static Pressure Contour Plot for Grid 2 at 15°

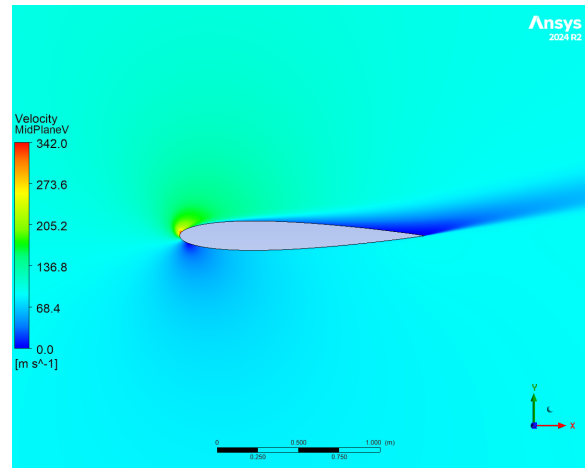


Figure 5.10: Velocity Contour Plot for Grid 2 at 15°

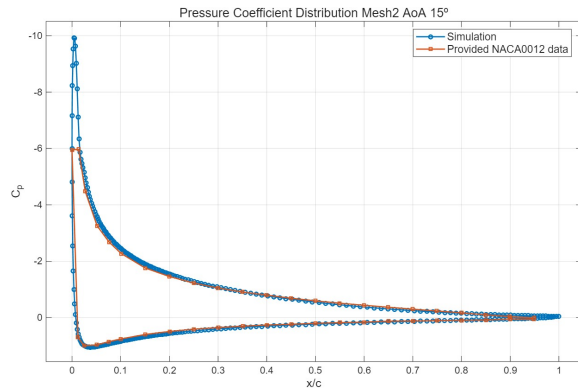
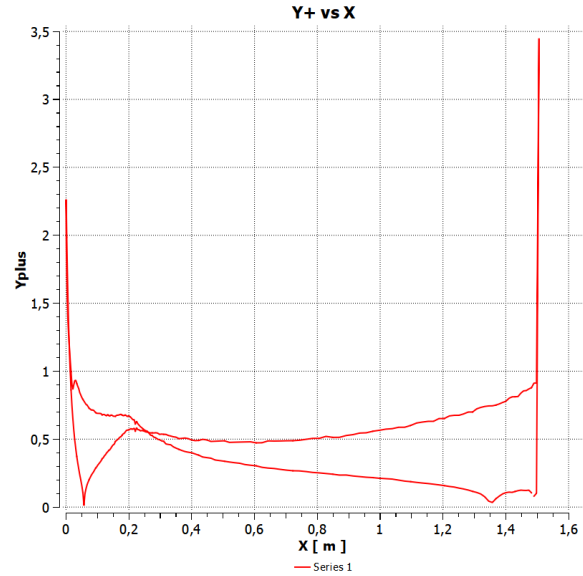
5.2.3. Non-Dimensional Numbers

The increase in pressure differences between the upper and lower surface can be spotted by the increase in C_L shown in Table 5.2, but C_D more than doubles. This observation is consistent with the onset of flow separation at higher angles of attack. The maximum Mach number also increases, as local acceleration over the upper surface increases as shown in the velocity contour plot above, reaching transonic conditions. This stronger acceleration shows that the upper surface now exhibits a stronger and narrower suction peak, followed by quicker pressure recovery revealing flow separation, shown in Figure 5.11. The reduced pressure on

Table 5.2: Non-Dimensional Coefficients of Grid 2 at 15°

Geometry	M_{max}	C_D	C_L
Airfoil Grid 2 at 15°	1.2348	6.2848e-02	1.3750

the lower surface suggests that the airfoil is approaching stall conditions. The correlation with the reference data is excellent, although the difference in the leading-edge suction peak arises from coarse data points, as explained before.

**Figure 5.11:** C_p Distribution of Airfoil Grid 2 and Reference Data at 15°**Figure 5.12:** y^+ Distribution of Grid 2 at 15°

Finally, the y^+ value along the airfoil's chord is shown in Figure 5.12. The leading and trailing edge peaks are now stronger than before indicating low separation. This is still consistent with the interpretations taken from the plots above. The rest of the chord is still at a y^+ value below 1, so the mesh still accurately resolves near-wall effects.

6 Verification

To verify that our simulation indeed solves what we want it to solve, two verification process are done. The first one is to repeat the simulation for 15° angle of attack with a first-order upwind scheme (UDS) to quantify numerical errors. The second one is to quantify the effect of the turbulence model, by repeating the simulation for 15° angle of attack with a turbulence model that can better account for anisotropic turbulence.

6.1. First-Order Upwind Scheme

By using a first-order upwind scheme we introduce more numerical diffusion, making it less accurate. Comparing it to the second-order scheme will show how sensitive the previous solutions is to discretizations errors. If the results don't change much, it will mean that the simulations are numerically stable and accurate. If the results change significantly, then the flow might be under-resolved or numerically dissipative.

6.1.1. Solver Residuals

The following graphs show the solver residuals:

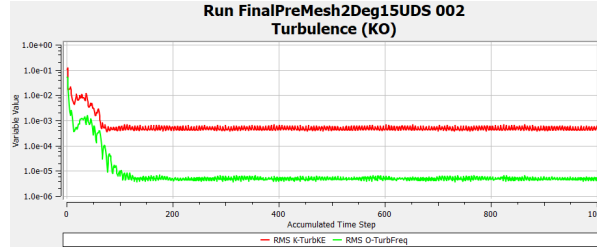


Figure 6.1: Solver Convergence for UDS Scheme

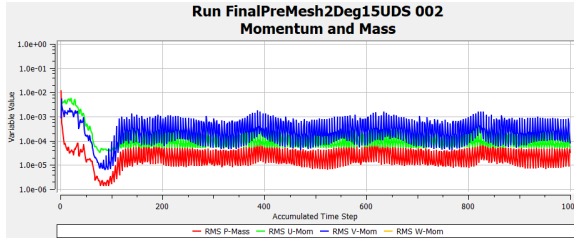


Figure 6.2: Solver Residuals for Grid 2 at 15° for UDS

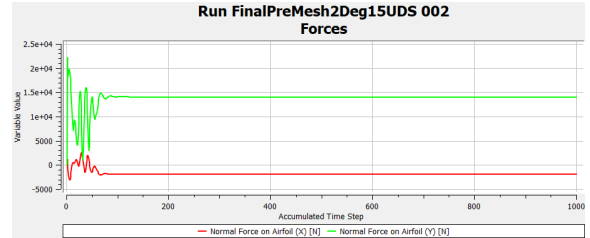


Figure 6.3: Force Convergence for Grid 2 at 15° for UDS

All of the variables converge and are below 10^{-3} , so the convergence quality is numerically good, although physical accuracy is limited by the first-order scheme. This is further discussed below.

6.1.2. Flow Field

The contour plots for the upwind scheme are smoother and more diffused. Figure 6.4 shows a weaker suction peak and it shows less pressure differences between the upper and lower surfaces compared to the non-upwind plot. Figure 6.5 also shows the velocity contours. Due to artificial diffusion from the upwind scheme, the low velocity separated flow spreads more.

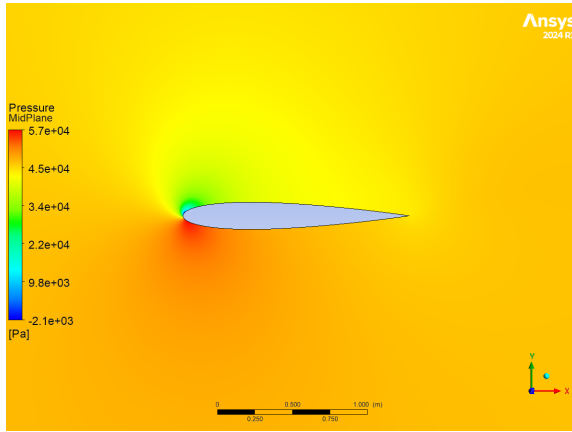


Figure 6.4: Static Pressure Contour Plot for Grid 2 at 15° for UDS

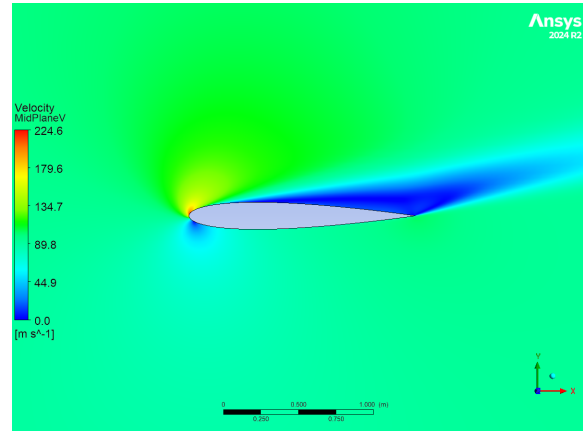


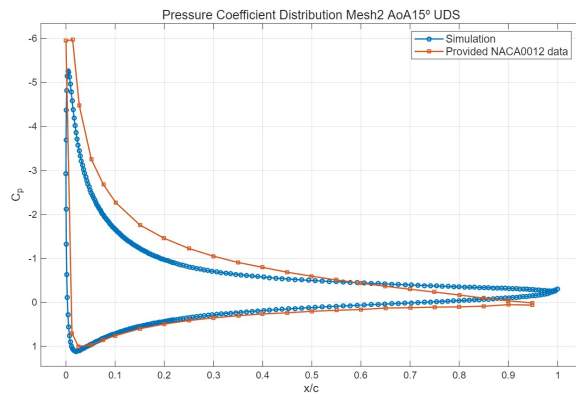
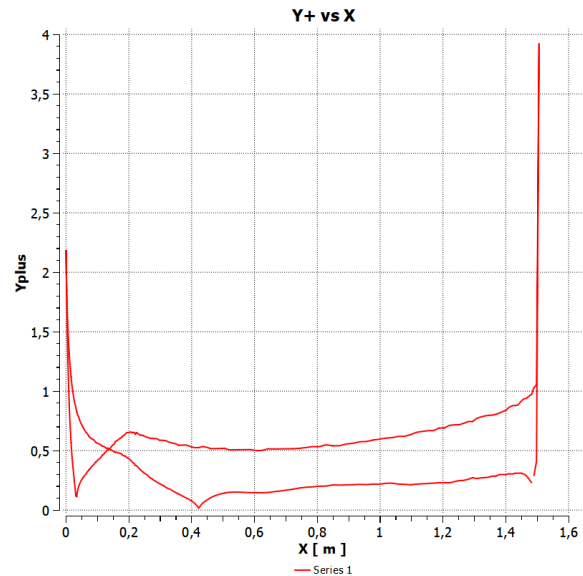
Figure 6.5: Velocity Contour Plot for Grid 2 at 15° for UDS

6.1.3. Non-Dimensional Numbers

The first-order upwind scheme introduces numerical diffusion which smooths out pressure and velocity gradients. This can be seen Figure 6.6. This smoothing also affects the aerodynamic coefficients. From Table 6.1, it can be seen how M_{max} and C_L are lower. Furthermore, C_D is higher because UDS introduces artificial numerical dissipation which acts as an extra drag force. Finally, Figure 6.7 shows the y^+ value for the first-order upwind scheme. As numerical diffusion smears gradients, there is an earlier apparent separation that elevates the y^+ value. This is seen in the rise towards the trailing edge.

Table 6.1: Non-Dimensional Coefficients of Grid 2 at 15° for UDS

Geometry	M_{max}	C_D	C_L
Airfoil Grid 2 at 15° UDS	0.7475	1.3479e-01	1.0076

Figure 6.6: C_p Distribution of Airfoil Grid 2 and Reference Data at 15° for UDSFigure 6.7: y^+ Distribution of Grid 2 at 15° for UDS

6.2. Turbulence Model

The Explicit Algebraic Reynolds Stress Model (EARSM) is now used. By using this different turbulence model that better captures anisotropic turbulence, it can be assessed how the choice of turbulence model affects the results.

6.2.1. Solver Residuals

For the turbulence convergence, the output of the original 15° was used as the initial condition for this solver to increase convergence rate. This can be seen in Figure 6.9, where the solver makes a jump upwards at 1200 iterations due to the change of the turbulence model. Additionally, once it stabilized, the time-step was changed to a local time-step with a ratio of three at 1800 iterations. From there it can be seen how the residuals decrease and converge below 10^{-4} . The same process was done for Figure 6.9 and Figure 4.6.

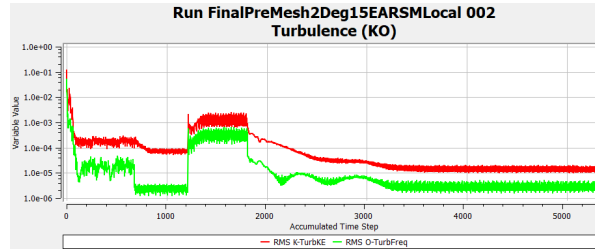


Figure 6.8: Turbulence Residuals for EARS M

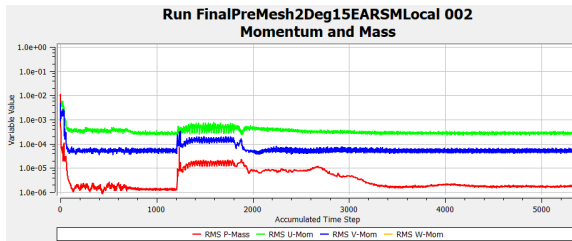


Figure 6.9: Solver Residuals for Grid 2 at 15° for EARS M

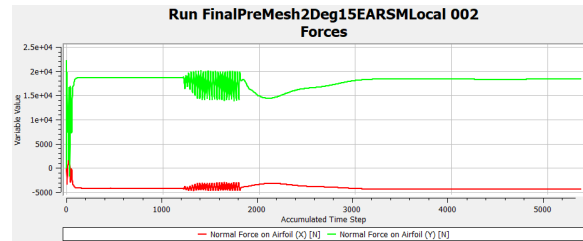


Figure 6.10: Force Convergence for Grid 2 at 15° for EARS M

6.2.2. Flow Field

The low pressure region on the upper surface from Figure 6.11 is sharper and stronger compared to the cases before. The pressure recovery looks smoother, indicating a more controlled flow separation. This is mirrored in Figure 6.12, where the high velocity region over the upper surface extends further downstream implying later separation. The wake region is thinner showing less numerical diffusion

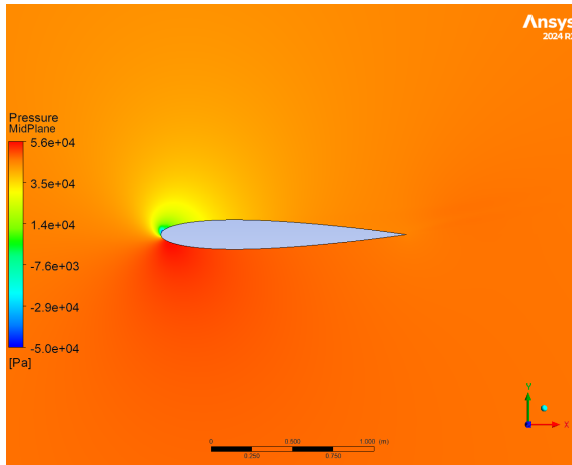


Figure 6.11: Static Pressure Contour Plot for Grid 2 at 15° for EARS M

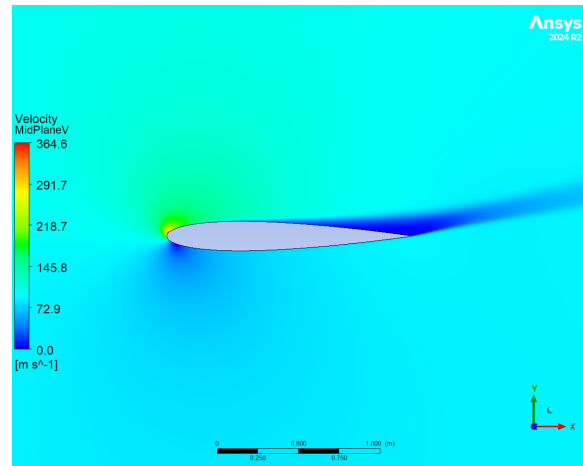


Figure 6.12: Velocity Contour Plot for Grid 2 at 15° for EARS M

6.2.3. Non-Dimensional Numbers

This turbulent model reduces overly large eddy viscosity in adverse pressure regions, which delays separation. This results in stronger suction peaks and drag reduction. It is reflected in Table 6.2, where M_{max} and C_L is

increased while C_D is decreased. The high suction peak and gradual pressure recovery toward the trailing edge is visualized in Figure 6.13. Finally, Figure 6.14 shows the y^+ value for the new turbulence model. There is also an increase of the y^+ value towards the trailing edge but it is smaller than for the previous turbulence model, due to delayed separation/

Table 6.2: Non-Dimensional Coefficients of Grid 2 at 15° for EARSM

Geometry	M_{max}	C_D	C_L
Airfoil Grid 2 at 15° EARSM	1.3366	4.3814e-02	1.3611

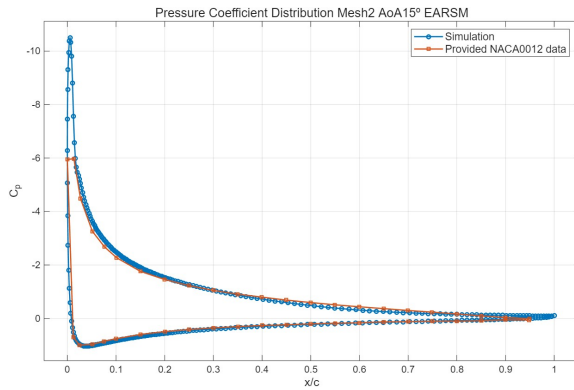


Figure 6.13: C_p Distribution of Airfoil Grid 2 and Reference Data at 15° for EARSM

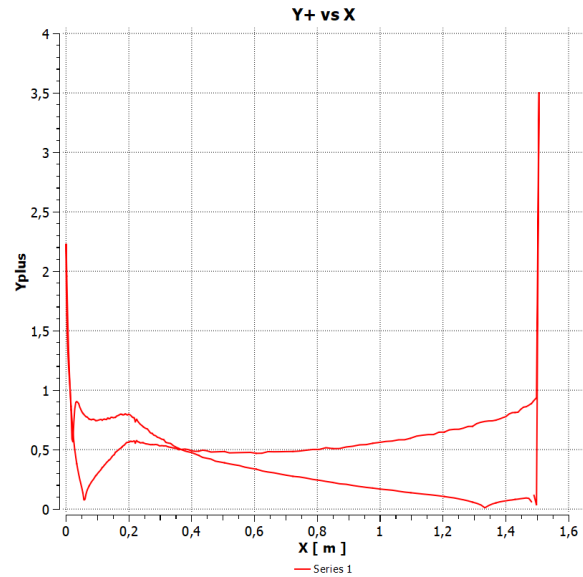


Figure 6.14: y^+ Distribution of Grid 2 at 15° for EARSM

7 Conclusion

This study successfully analyzed the NACA0012 airfoil aerodynamics across three angles of attack using 2D steady-state RANS simulations. Mesh refinement was done for the second mesh to achieve $y^+ < 1$ proved essential for accurate near-wall flow resolution, yielding the expected near-zero lift at 0° for the symmetric airfoil.

The simulations captured the transition from attached to separated flow as angle of attack increased, therefore increasing lift and drag coefficients and maximum Mach number.

The verification studies revealed critical insights: the first-order upwind scheme underpredicted lift and overpredicted drag due to excessive numerical dissipation, demonstrating that numerical errors can completely overshadow physical phenomena. The EARSM turbulence model reduced drag compared to $k - \omega$ by better capturing flow separation, highlighting the significant impact of turbulence modeling on separated flows.

All results showed excellent agreement with experimental reference data. Future work could include a better mesh refinement or maybe running unsteady simulations to capture vortex shedding at post-stall conditions at the different angles of attack.